ADVANCING AI 2026

BUILD WHAT'S NEXT

Accelerating LLM Inference on AMD GPUs with AMD ATOM - AiTer Optimized Model

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WORKSHOP PRE- REQUISITE

Make sure you’re enrolled in the AMD Developer Program

You must complete this step to use the Instinct GPUs and participate in this workshop.

Additional Developer Program Benefits:

- Complimentary GPU Credits
- One-month DeepLearning.ai Pro membership
- Access to AMD AI Academy courses
- Dedicated technical support
- Rewards and recognitions

https://developer.amd.com/ai-developer-program/

Agenda

1 What Is ATOM

Definition, mission, and architecture role

2 ATOM Core Capabilities

Key features in scheduling, kernels, precision, and scaling

3 Open Ecosystem Compatibility

Integration with vLLM and SGLang through the shared ATOM core

4 ATOM Hands-on Exercise

Using ATOM LLM engine on AMD Instinct GPUs

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AMD AI Software Stack Overview

ATOM / AITER / ROCm Each Layer Optimized, All Layers Aligned.

ATOM

github.com/ROCm/ATOM

Inference Engine

- OpenAI-compatible API server
- Scheduler + KV cache manager
- Model implementations
- Torch.compile
- HipGraph
- TP / DP / EP parallelism
- MTP speculative decoding
- vLLM & SGLang plugin

AITER

github.com/ROCm/aiter

Fast Kernel Library

- ASM, FlyDSL, Triton backends
- Flash Attention (MHA & MLA)
- Paged Attention (ASM)
- GEMM: FP8, MXFP4, INT8, INT4
- Fused MoE execution
- RMSNorm, SiLU, RoPE fusions
- Quantization-aware ops

ROCm

github.com/ROCm/ROCm

AI Software Stack

- Open-source AMD accelerator platform
- Runtime + compiler + driver ecosystem
- HIP / RCCL / MIOpen / rocBLAS
- Profiling and debugging toolchain
- Foundation for ATOM / AITER / MoRI

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ATOM×AITER: Co-Designed Modeling for AMD GPUs

Integrated optimization path

01 Aggressive Operator Fusion

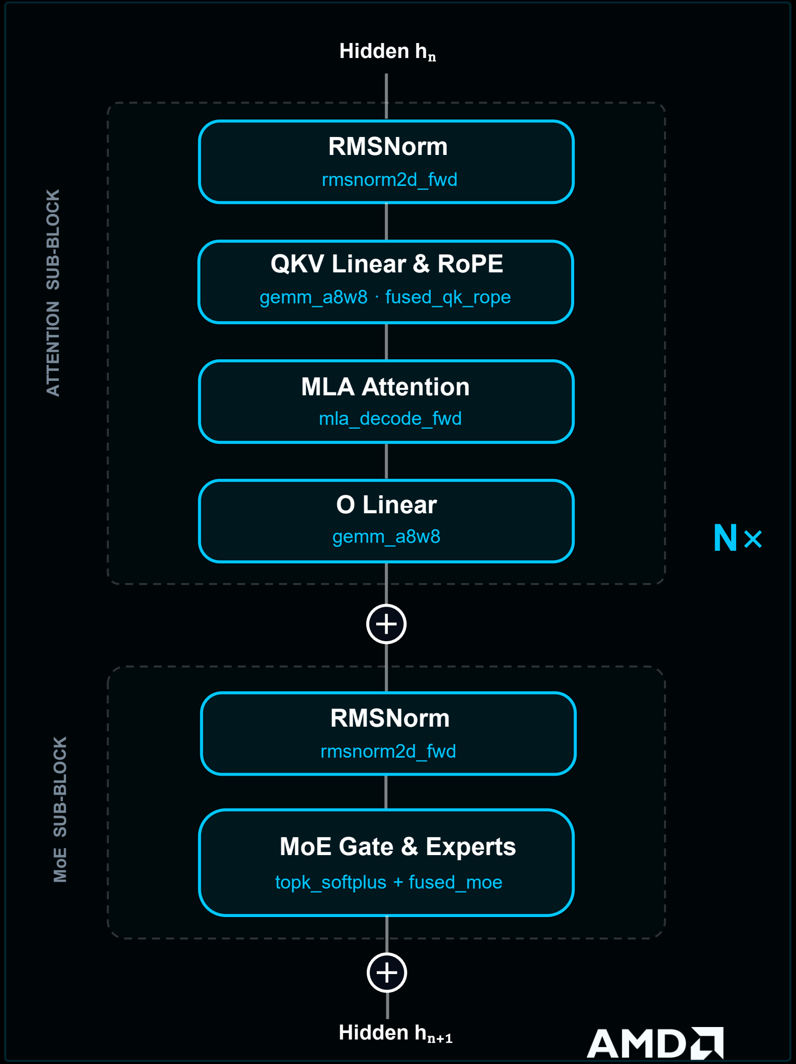
Many ops collapsed into one launch — fused_qk_rope_cache wraps norm, RoPE, and KV-cache write together; fused_moe wraps gate, GEMM, and dequant. Fewer launches, less HBM round-trips.

02 Critical Ops Hand-Tuned, Broadly Generalizable

Only critical kernels are hand-tuned for CDNA MFMA tensor cores. FlyDSL defines tile and LDS layout; Triton provides fast paths to keep performance portable across model families.

03 Per-Model Parallelism Strategies

TP / DP / EP picked per architecture — Dense models use TP+DP, MoE models use TP+EP with MORI all-to-all, long-context uses DP attention. One engine, many topologies.



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ATOM Architecture

Two Deploy Modes: Performance Max & Ecosystem Compatible

OpenAI-Compatible API

vLLM Scheduler | **ATOM EngineCore Scheduler** | **SGLang Scheduler**

vLLM ModelRunner | **ATOM ModelRunner** | **SGLang ModelRunner**

SHARED **ATOM Models + Model Ops**

DeepSeek · Qwen · Llama · PagedAttention · FusedMoE · Sampler

MoRI **AITER** **RCCL**

AMD Instinct™ GPU · ROCm Stack

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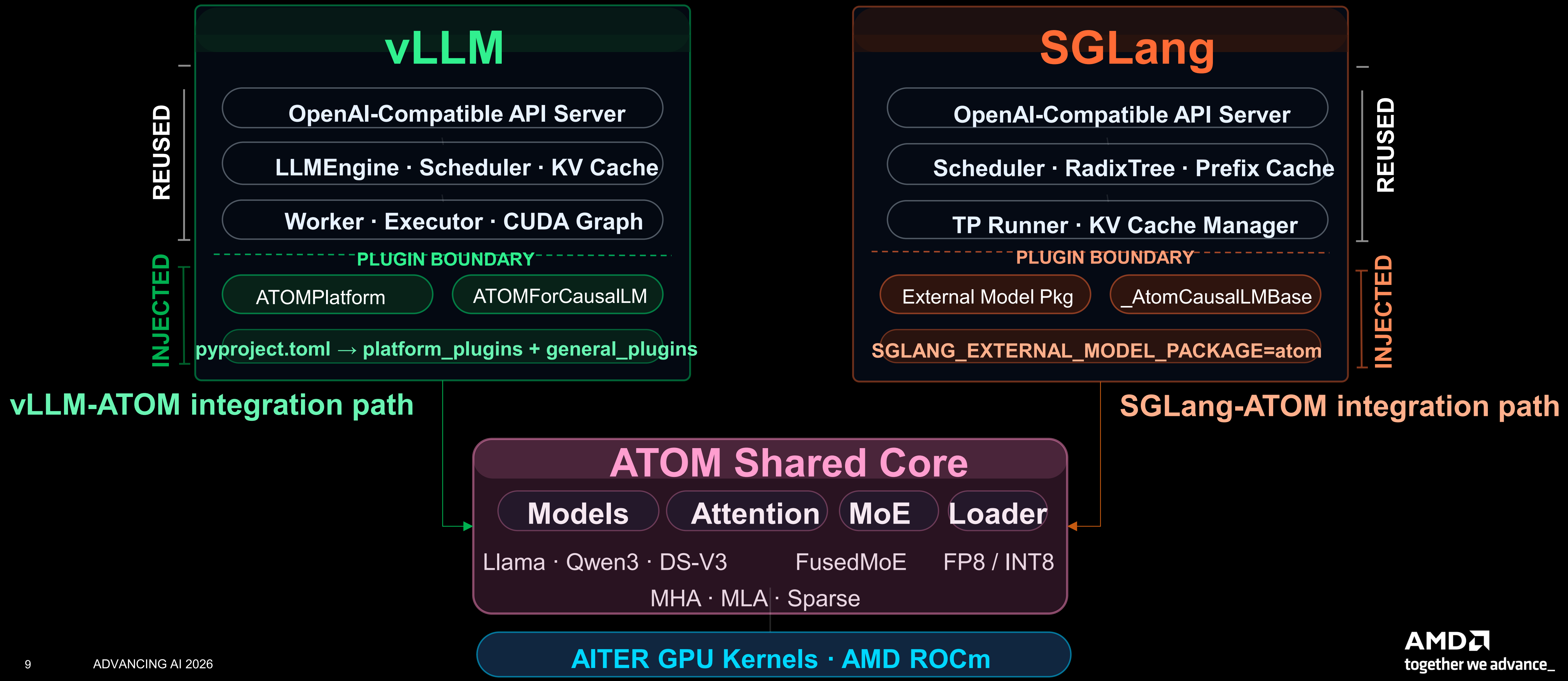
ATOM-Optimized Models via Official Interfaces

ATOM is a reference impl. with full compatibility with SGLang and vLLM



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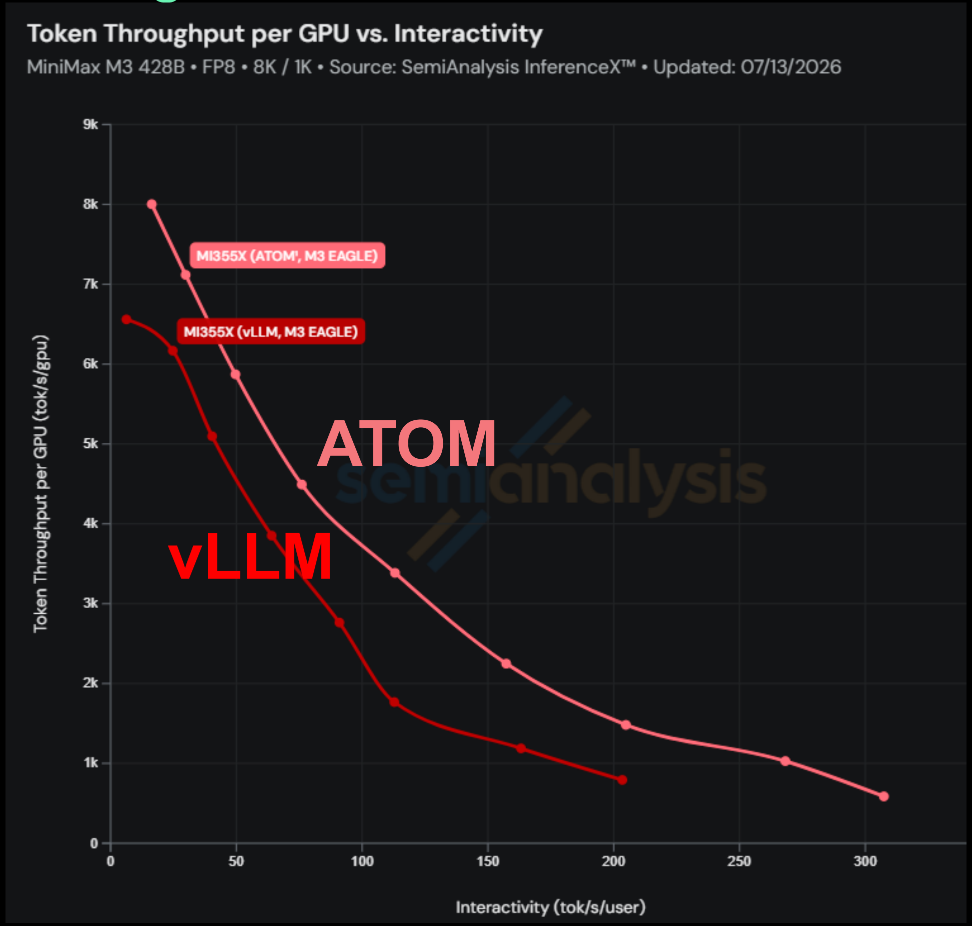
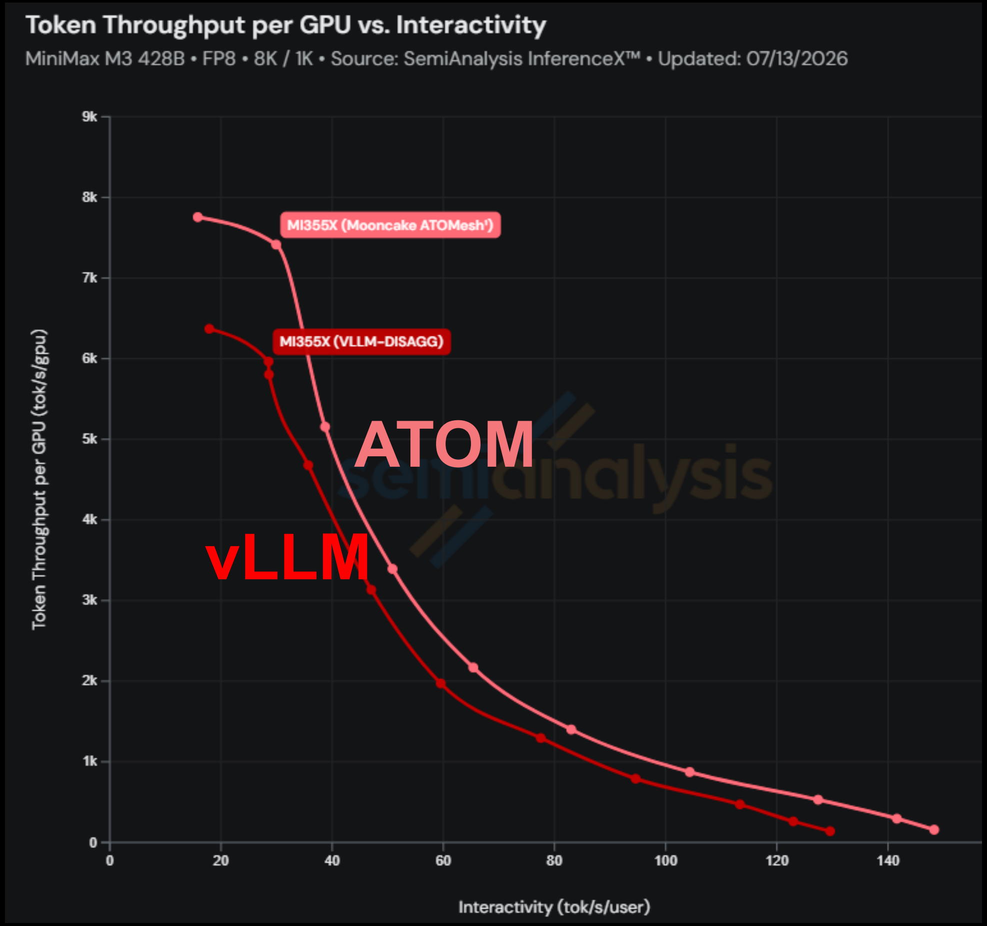
ATOM model performance

ATOM new model performance Minimax-M3

Single Node



Multi Node



https://inferencex.semianalysis.com/inference

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ATOM Hands-on Exercise

Launch a Jupyter Notebook:

https://notebooks.amd.com/aai

PART1. ATOM performance with MTP

- Multi-Token Prediction Prediction

ATOM Inference Engine

- ATOM basic Usage
- Example: LLM Performance metrics with MTP

PART2. Vibe coding with ATOM Claude

- Vibe coding with ATOM-Claude

ATOM Inference Engine + Claude

- ATOM-Claude Vibe coding
- Example: GeMM tuning

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3:00PM Accelerating LLM Inference on AMD GPUs with AMD ATOM

Complete this survey for a chance to bring home AMD Hardware today!

AMD Dev Quest Code

Developer Resources

AMD AI Developer Program

Gain access to a core set of resources like cloud credits, tools, training, and community support

Join the AI DevMaster Hackathon

Deploy Agentic AI, Physical AI and Multi-Modality on Radeon GPUs
Submissions close Aug 6

AMD AI Playbooks

Access step-by-step guides to run AI workloads on AMD hardware

AMD Developer Discord

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